

MACEIO, Brazil

WMO: 829930

Lat: 9.511S

Lon: 35.792W

Elev: 118

StdP: 99.92

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.5	19.0	16.8	12.2	23.7	17.5	12.7	23.4	8.5	25.6	8.0	25.7	1.5	320	0.306

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	7.9	33.0	25.5	32.2	25.1	31.8	25.0	26.8	30.3	26.4	29.8	26.1	29.5	5.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	21.8	28.5	25.6	21.2	28.1	25.2	20.6	27.6	85.4	30.4	83.2	29.9	81.5	29.7	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.4	6.8	DB	17.2	35.6	1.0	1.2	16.5	36.4	15.9	37.2	15.4	37.9	14.7	38.7	(4)
			WB	16.6	28.2	1.1	1.0	15.7	29.0	15.1	29.6	14.4	30.2	13.6	30.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.6	26.8	27.1	27.3	26.7	25.5	24.4	23.7	23.6	24.3	25.4	26.3	26.8
	DBStd	1.71	1.27	1.17	1.12	1.03	1.09	1.12	1.17	1.08	1.06	1.03	1.10	1.05
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5708	522	478	536	499	479	433	424	422	429	476	488	522
	CDD18.3	2667	263	244	278	249	221	183	166	164	179	218	238	263
	CDH23.3	20566	2383	2288	2647	2130	1511	978	744	810	1118	1606	1998	2354
	CDH26.7	6382	860	831	986	691	369	152	78	101	223	481	727	884
Wind	WSAvg	3.0	3.3	3.1	2.9	2.6	2.5	2.6	2.8	2.9	3.1	3.4	3.6	3.6
Precipitation	PrecAvg	1495	58	75	116	182	232	258	215	140	97	63	29	30
	PrecMax	2024	201	238	290	484	597	494	412	299	307	235	148	93
	PrecMin	836	0	9	36	45	44	86	74	42	15	1	0	3
	PrecStd	320	51	50	59	93	115	112	80	59	74	58	36	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	33.4	33.2	33.0	31.2	29.9	29.1	29.8	30.7	31.9	33.0	33.9
		MCWB	26.1	25.8	25.7	25.6	25.1	24.7	24.3	23.9	24.8	24.2	25.4	25.5
	2%	DB	32.3	32.2	32.4	31.9	30.2	29.0	28.2	28.4	29.3	30.9	31.9	32.2
		MCWB	25.3	25.3	25.3	25.2	24.8	24.2	23.7	23.3	23.7	24.3	24.8	24.9
	5%	DB	31.2	31.8	31.9	31.0	29.7	28.2	27.7	27.9	28.9	30.0	31.1	31.5
		MCWB	24.6	25.1	25.2	25.1	24.6	23.7	23.3	22.9	23.4	23.9	24.3	24.6
	10%	DB	30.7	30.9	31.1	30.1	28.9	27.7	26.9	27.0	28.0	29.1	30.2	30.9
		MCWB	24.4	24.7	24.9	25.0	24.4	23.5	22.8	22.5	23.0	23.6	24.0	24.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	27.1	27.5	27.2	26.7	26.0	25.8	25.4	25.7	26.0	26.6	26.8
		MCDB	31.3	30.6	31.0	29.9	29.2	28.2	27.6	27.3	28.9	29.4	30.7	31.0
	2%	WB	26.4	26.4	26.6	26.5	26.0	25.2	25.0	24.6	25.0	25.2	25.9	26.1
		MCDB	30.0	29.9	30.1	29.5	28.7	27.5	26.9	26.8	27.8	28.8	29.8	30.1
	5%	WB	25.9	26.0	26.2	26.1	25.5	24.7	24.4	24.0	24.4	24.7	25.3	25.6
		MCDB	29.3	29.5	29.6	29.1	28.1	26.9	26.3	26.4	27.2	28.2	29.2	29.5
	10%	WB	25.5	25.6	25.8	25.7	25.0	24.3	23.7	23.5	23.8	24.3	24.8	25.1
		MCDB	28.8	29.0	29.2	28.6	27.5	26.4	25.6	25.8	26.6	27.7	28.5	28.8
Mean Daily Temperature Range	5% DB	MDBR	8.0	7.9	7.9	7.3	6.8	6.5	6.5	7.2	7.8	8.4	8.7	8.6
		MCDBR	8.9	8.8	8.8	8.3	7.9	7.2	7.3	8.2	8.8	9.2	9.5	9.4
		MCWBR	3.0	2.9	3.0	3.1	3.5	3.5	3.7	3.9	4.0	3.7	3.4	3.2
	5% WB	MCDBR	8.1	7.9	7.8	7.3	6.8	6.3	6.4	7.1	7.8	8.4	8.6	8.6
	MCWBR	3.5	3.3	3.3	3.5	3.6	3.6	3.9	4.2	4.2	3.9	3.7	3.6	
Clear-Sky Solar Irradiance	taub	0.386	0.386	0.386	0.378	0.374	0.373	0.371	0.382	0.410	0.408	0.393	0.396	
	taud	2.510	2.519	2.524	2.560	2.577	2.563	2.556	2.489	2.385	2.403	2.458	2.462	
	Ebn at Noon	954	950	934	912	887	873	882	896	898	918	941	943	
	Edh at Noon	114	112	110	101	94	93	95	107	124	125	119	119	
All-Sky Solar Radiation	RadAvg	6.41	6.31	6.21	5.65	4.99	4.56	4.77	5.31	5.95	6.34	6.79	6.70	
	RadStd	0.48	0.26	0.27	0.34	0.40	0.29	0.27	0.27	0.30	0.27	0.32	0.26	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		-0.41	-0.67	-1.15	N/A	-0.68	-0.74	N/A	N/A	-156	-155
Regional (1 neighbor)		-0.64	-0.52	-0.93	-0.79	-0.66	-0.60	N/A	N/A	-264	-266

Nomenclature: See separate page